

Unit 8, Lesson 1: Proving Properties of Parallelograms



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can define a parallelogram.
- ☐ I can prove opposite sides and opposite angles of a parallelogram are congruent.
- ☐ I can prove consecutive angles of a parallelogram are supplementary.
- ☐ I can prove diagonals of a parallelogram bisect each other.



8.1.1 Warm-Up: Cartographer

Dave Imus is a cartographer whose maps have won many awards. Dave sees the world through understanding places. Dave's maps are considered art.

1. Have you ever used a paper map?

2. Since you can't put everything on a map, what three places do you think should be on the map of Florida?





8.1.2 Collaboration: Parallel Lines and Four-Sided Figures

1. On a piece of patty paper, draw one set of parallel lines. Overlap your patty paper with a partner's parallel lines to create an enclosed figure. Each partner should trace the enclosed shape onto a new piece of patty paper.
 - a. Classify the enclosed shape.
 - b. Justify your classification.
 - c. Ask two classmates how they classified the figure they created. Record their classification and write your summary of their justification. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Classification	My Summary of Their Justification	Initials



8.1.3 Exploration: Properties of Parallelograms



A quadrilateral containing two pairs of parallel sides is a **parallelogram**. A member of the following shape classes: polygons, quadrilaterals, trapezoids. Sub-classes include rectangles, rhombi, and squares.

1. Using a ruler, measure the pairs of opposite sides in each parallelogram you and your partner created.
 - a. What do you notice?
 - b. Make a conjecture about the measures of opposite sides of a parallelogram.
2. Using a ruler, draw the diagonals of one of the parallelograms.
 - a. Determine the lengths of each diagonal.
 - b. Mark the point where the two diagonals intersect.

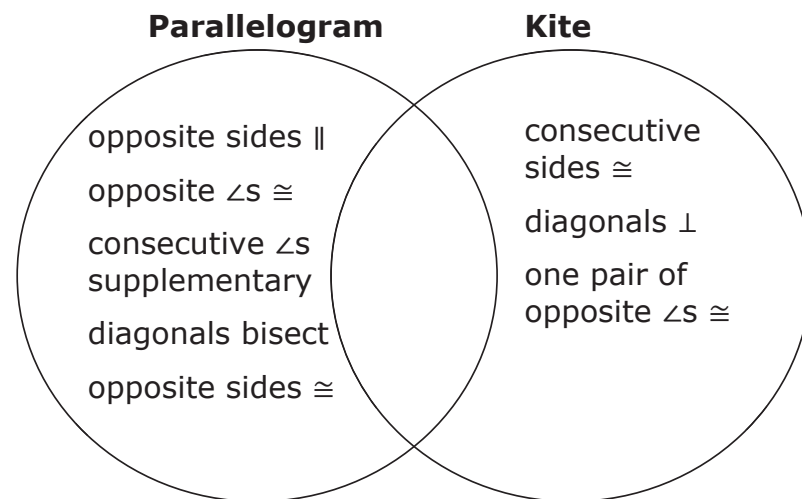
- c. What do you notice about the point of intersection and its relationship to each diagonal?
- d. Repeat the same steps to draw and measure the diagonals in your second parallelogram.
- e. Make a conjecture about the diagonals of a parallelogram.
3. Using a protractor, measure the pairs of opposite interior angles of your parallelograms.
- a. What do you notice?
- b. Make a conjecture about the measures of opposite angles of a parallelogram.
- c. Make a conjecture about the measures of consecutive angles of a parallelogram.
4. Based on your conjectures, select all of the statements about a parallelogram that are true.
- ☐ A parallelogram has only one set of opposite parallel sides.
 - ☐ A parallelogram has two sets of opposite parallel sides.
 - ☐ A parallelogram has only one set of opposite congruent sides.
 - ☐ A parallelogram has two sets of opposite congruent sides.
 - ☐ Consecutive angles of a parallelogram are congruent.
 - ☐ Consecutive angles of a parallelogram are supplementary.
 - ☐ Opposite angles of a parallelogram are congruent.
 - ☐ Opposite angles of a parallelogram are supplementary.
 - ☐ Diagonals of a parallelogram bisect each other.



5. For each true statement in the previous problem, draw and name a parallelogram. Using the information in the statement, mark the diagram and write a mathematical statement. The first row has been completed.

Statement	Drawing	Given
A parallelogram has two sets of opposite parallel sides.		$\overline{AD} \parallel \overline{BC}$ $\overline{AB} \parallel \overline{DC}$

6. The Venn diagram shows the properties of a parallelogram and another quadrilateral called a kite. Discuss with your partner why a kite is not considered a parallelogram.



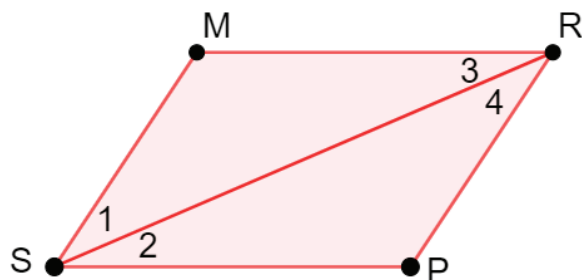
Properties of Parallelograms

- Opposite sides of a parallelogram are congruent.
- Opposite angles of a parallelogram are congruent.
- Consecutive interior angles of a parallelogram are supplementary.
- The diagonals of a parallelogram bisect each other.



8.1.4 Guided Instruction: Proofs with Parallelograms

1. $MRPS$ is shown.



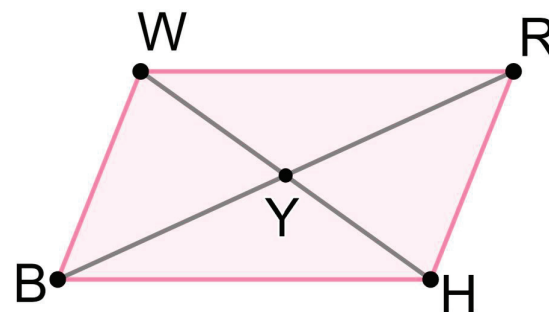
Given: $MRPS$ is a parallelogram

Prove: $\overline{MR} \cong \overline{PS}$ and $\overline{MS} \cong \overline{PR}$

Complete the following two-column proof.

Statement	Reason
1.	1. Given
2.	2.
3. $\angle 1 \cong \angle 4, \angle 2 \cong \angle 3$	3.
4. $\overline{SR} \cong \overline{SR}$	4. Reflexive property of congruence
5. $\triangle MSR \cong \triangle PRS$	5. ASA congruence theorem
6.	6. CPCTC

2. $WRHB$ with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y is shown.



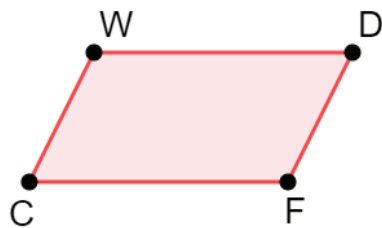
Given: $WRHB$ is a parallelogram with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y

Prove: $\overline{WH}$ bisects $\overline{BR}$
 $\overline{BR}$ bisects $\overline{WH}$

Complete the following two-column proof.

Statement	Reason
1.	1. Given
2.	2.
3. $\overline{RH} \cong \overline{BW}; \overline{WR} \cong \overline{HB}$	3.
4. $\angle WRB \cong \angle HBR;$ $\angle BHW \cong \angle RWH$	4.
5. $\triangle BYH \cong \triangle RYW$	5. ASA congruence theorem
6.	6. CPCTC
7. $BY = YR$ and $WY = YH$	7.
8.	8. Definition of a segment bisector

3. Complete the following flowchart proof. Statements go in the boxes and reasons go below the boxes.

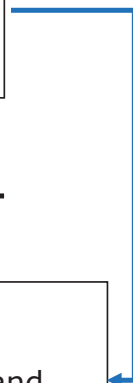


Given: $WDFC$ is a parallelogram.

Prove: $\angle W$ is supplementary to $\angle D$,
 $\angle D$ is supplementary to $\angle F$,
 $\angle F$ is supplementary to $\angle C$, and
 $\angle C$ is supplementary to $\angle W$



Given



$\angle W$ is supplementary to $\angle D$,
 $\angle D$ is supplementary to $\angle F$,
 $\angle F$ is supplementary to $\angle C$, and
 $\angle C$ is supplementary to $\angle W$



8.1.5 Wrap-Up: Cool Down

1. Write down one fact you knew about parallelograms before the lesson.
2. Write down one fact you learned about parallelograms during the lesson.